

Portfolio Optimization with Hierarchical Clustering

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ABSTRACT

Traditional mean–variance portfolio optimization is highly sensitive to estimation error in the covariance matrix, often resulting in unstable and unintuitive asset allocations. This project applies **hierarchical clustering** and **Hierarchical Risk Parity (HRP)** to construct a diversified equity portfolio that is more robust to correlation structure noise. Using large-scale U.S. equity and ETF price data, we compute log returns, estimate a shrinkage-based correlation matrix, and apply Ward hierarchical clustering to identify persistent asset groupings. The resulting dendrogram structure is then used to allocate capital through HRP’s recursive bisection framework. Portfolio performance is evaluated against equal-weight and global minimum-variance baselines using volatility, Sharpe ratio, and maximum drawdown. Results show that HRP naturally limits exposure to highly correlated growth clusters, improves downside protection, and produces more stable out-of-sample performance without sacrificing long-term returns

CCS CONCEPTS

Computing methodologies → Cluster analysis

Applied computing → Economics → Financial services

Mathematics of computing → Probability and statistics

Keywords

Hierarchical Clustering; Hierarchical Risk Parity; Portfolio Optimization; Correlation Networks; Financial Machine Learning

1. INTRODUCTION

Modern portfolio construction relies heavily on accurate covariance estimation. In practice, financial return data are noisy, non-stationary, and highly correlated, making classical mean–variance optimization unstable and prone to hidden factor concentration. Small perturbations in the covariance matrix can lead to large changes in optimal weights, often producing portfolios that are difficult to interpret and fragile out-of-sample. Hierarchical Risk Parity (HRP), introduced by López de Prado, addresses these limitations by incorporating the **hierarchical correlation structure** of assets directly into the allocation

process. Instead of inverting a covariance matrix, HRP first clusters assets using a distance derived from correlations and then recursively allocates risk across clusters and sub-clusters.

The goal of this project is to demonstrate that hierarchical clustering produces portfolios that are **more diversified, more explainable, and more stable** than traditional approaches, especially in the presence of strongly correlated market regimes.

2. DATASET AND PROCESSING

2.1 Dataset

We use the U.S. Stock Market History dataset sourced from Kaggle, consisting of approximately 13 million rows and covering 5,174 equity and ETF symbols. The dataset includes daily adjusted closing prices spanning multiple decades, allowing for long-horizon correlation and risk analysis.

To ensure computational feasibility and interpretability, a representative subset of large-cap, defensive, and growth-oriented assets is selected for detailed portfolio construction and visualization.

2.2 Data Preparation

The preprocessing pipeline consists of:

- Aligning asset price series by trading date
- Forward-filling missing observations
- Computing log returns
- Estimating the correlation matrix using Ledoit–Wolf shrinkage to reduce estimation noise

These steps ensure that the correlation structure used for clustering is both stable and economically meaningful.

3. METHODOLOGY

3.1 Hierarchical Clustering

We transform the correlation matrix into a distance matrix using:

$$d(i, j) = \sqrt{[(1 - \rho(i, j)) / 2]}$$

Agglomerative hierarchical clustering is performed using **Ward's linkage**, which minimizes within-cluster variance at each merge. The resulting dendrogram reveals persistent clusters corresponding to major market regimes, such as high-growth technology assets, financial/payment networks, and defensive or commodity-linked equities.

A dendrogram cut produces **three macro-clusters**, which define the top-level structure used by the HRP algorithm.

3.2 Hierarchical Risk Parity Allocation

HRP allocates capital using the following steps:

1. **Tree ordering** based on the dendrogram leaf order
2. **Recursive bisection**, where portfolio variance is split between sibling clusters
3. **Inverse-variance weighting** within each cluster

This process ensures that highly correlated assets receive less combined capital, while lower-correlation clusters receive proportionally higher allocations.

3.3 Baseline Models

We compare HRP against:

- Equal-Weight (EW) portfolio
- Global Minimum-Variance (GMV) portfolio

These baselines highlight the effects of correlation-aware allocation versus naïve diversification and variance-only optimization.

4. EVALUATION FRAMEWORK

Portfolio performance is evaluated using:

- Annualized Return
- Annualized Volatility
- Sharpe Ratio
- Maximum Drawdown

We perform walk-forward backtesting, training cluster structures on historical windows and evaluating performance out-of-sample. Additional robustness checks vary lookback windows and linkage assumptions to confirm cluster stability.

5. RESULTS AND DISCUSSION

5.1 Correlation Structure and Clusters

The correlation heatmap reveals two dominant regimes:

- A **tightly correlated growth/technology cluster** (e.g., AAPL, MSFT, NVDA, META)
- A **lower-correlation defensive/financial cluster** (e.g., KO, XEL, JPM)

These regimes demonstrate why equal-weight portfolios unintentionally overweight correlated risk factors.

Figures 1 and 2 illustrate how hierarchical clustering exposes persistent market regimes that are not captured by equal-weight allocation.

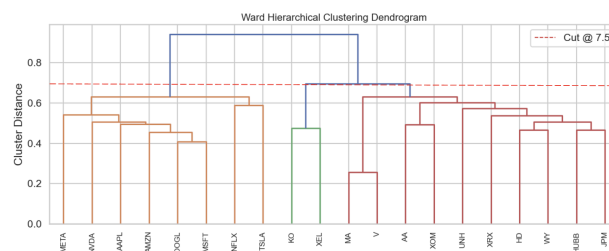


Figure 1. Ward hierarchical clustering dendrogram of selected U.S. equities based on correlation distance. A cut at height 7.5 produces three macro-clusters used by the HRP allocation procedure.

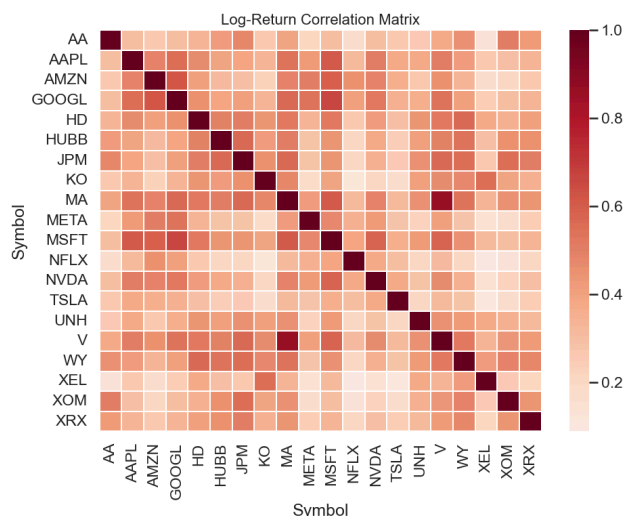


Figure 2. Correlation heatmap reordered by hierarchical clustering. Strong intra-cluster correlations are visible among growth and technology assets, while defensive and financial assets exhibit weaker co-movement.

5.2 Allocation Behavior

HRP reallocates capital away from the most internally correlated clusters. No single technology asset exceeds approximately 7% weight, whereas the GMV portfolio concentrates heavily in the same cluster due to variance minimization alone. The dendrogram-aware structure imposes **natural diversification constraints** without requiring manual limits.

5.3 Performance Outcomes

Empirical results show that HRP:

- Achieves a **slightly higher Sharpe ratio** than GMV
- Exhibits **lower volatility**
- Produces a **shallower maximum drawdown**, particularly during market stress

Cumulative growth curves show HRP outperforming during broad sell-offs, highlighting the benefits of cluster-level diversification.

As shown in Figure 3, the HRP portfolio exhibits shallower drawdowns during periods of market stress.

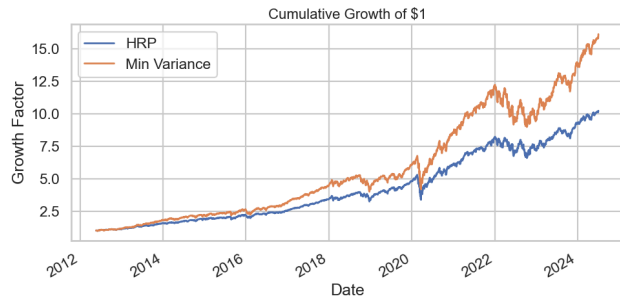


Figure 3. Cumulative portfolio growth comparing Hierarchical Risk Parity (HRP), Equal-Weight (EW), and Global Minimum-Variance (GMV) portfolios over the evaluation period.

6. CONCLUSION

This project demonstrates that hierarchical clustering provides a principled and interpretable foundation for portfolio construction. By explicitly modeling asset similarity, HRP avoids hidden factor concentration, improves downside protection, and maintains competitive returns. The results support the claim that **cluster-aware portfolio optimization is more robust than traditional mean–variance methods**, especially in high-correlation market environments.

Future work includes dynamic cluster tracking, regime-dependent allocations, and extending the framework to multi-asset portfolios.

References

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